
SafeSU User's Manual

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1 OVERVIEW

The SafeSU (Safe Statistics Unit) is an AHB slave capable of monitoring SoC events, enforce contention control, and identifying profiling errors on run-time. Figure "1.1" shows the structure of the unit. It is composed of an ahb wrapper (*ahb_wrapper.vhd*) that maps the SystemVerilog implementation into a VHDL module that can be instanced in SELENE and De-RISC SoCs. The SystemVerilog AHB interface (*pmu_ahb.sv*) offers support for a subset of AHB requests. This module also instances the interface agnostic SafeSU (*PMU_raw.sv*). The latter is used as the generator of the statistic unit. It generates the memory map and the instances for each of the features.

The main features are:

- **Self-test:** Allows to configure the counters' inputs to a fixed value bypassing the crossbar and ignoring the inputs. This mode allows for tests of the software and the unit under known conditions.
- **Crossbar:** Allows to route any input event to any counter.
- **Counters:** Group of simple counters with settable initial values and general control register.
- **Overflow:** Detection of overflow for counters, interrupt capable with dedicated interruption vector and per counter interrupt enable.
- **MCCU** (Maximum Contention Control Unit) : Contention control measures for each core. Interruption capable after a threshold of contention is exceeded. It accepts real contention signals or estimation through weights.
- **RDC** (Request Duration Counters): Provides measures of the pulse length of a given input signal (watermark). It can be used to determine maximum latency and cycles of uninterrupted contentions. Each of the counters can trigger an interrupt at a user-defined threshold.

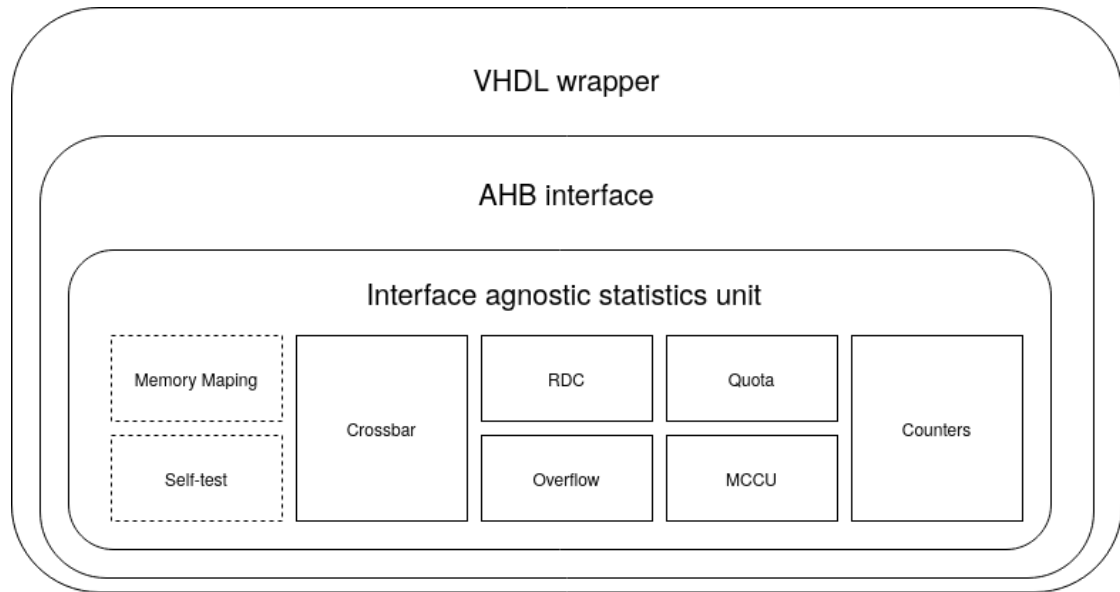


Figure 1.1: Block diagram statistics unit structure

2 OPERATION

2.1 GENERAL

The SafeSU attaches to a 128-bit wide AHB bus but only supports Single burst 32-bit accesses. AHB Lock accesses and protection control are not implemented.

The unit is compatible with GRLIB *plug&play*(P&P). Vendor and device's default configuration values are "BSC" and "AHB Performance Monitoring Unit" respectively. Regardless, release *v3.2.4* of GRMON, the unit may appear as "Unknown device".

2.2 DEFAULT INPUT EVENTS

In its default implementation, the unit provides up to 256 input events. Each one of the events can be routed to any mechanism of the module through the crossbar.

Table 2.1 shows the inputs and mapping to the crossbar input for the current SELENE release. The number of signals and arrangements may change in the upcoming versions.

Table 2.1: SELENE GPL input events. 256 inputs, 6 cores

Index	Type	Source	Description
0	Debug	local	Constant HIGH, used for debug purposes or clock cycles
1	Debug	local	Constant LOW, used for debug purposes
2	Pulse	Core 0	Instruction count pipeline 0
3	Pulse	Core 0	Instruction count pipeline 1
4	Pulse	Core 0	Instruction cache miss
5	Pulse	Core 0	Instruction TLB miss
6	Pulse	Core 0	Data cache L1 miss
7	Pulse	Core 0	Data TLB miss
8	Pulse	Core 0	Branch predictor miss
9	Pulse	Core 1	Instruction count pipeline 0
10	Pulse	Core 1	Instruction count pipeline 1
11	Pulse	Core 1	Instruction cache miss
12	Pulse	Core 1	Instruction TLB miss
13	Pulse	Core 1	Data cache L1 miss
14	Pulse	Core 1	Data TLB miss
15	Pulse	Core 1	Branch predictor miss
16	Pulse	Core 2	Instruction count pipeline 0
17	Pulse	Core 2	Instruction count pipeline 1
18	Pulse	Core 2	Instruction cache miss
19	Pulse	Core 2	Instruction TLB miss
20	Pulse	Core 2	Data cache L1 miss
21	Pulse	Core 2	Data TLB miss
22	Pulse	Core 2	Branch predictor miss
23	Pulse	Core 3	Instruction count pipeline 0
24	Pulse	Core 3	Instruction count pipeline 1
25	Pulse	Core 3	Instruction cache miss
26	Pulse	Core 3	Instruction TLB miss
27	Pulse	Core 3	Data cache L1 miss
28	Pulse	Core 3	Data TLB miss
29	Pulse	Core 3	Branch predictor miss
30	Pulse	Core 4	Instruction count pipeline 0
31	Pulse	Core 4	Instruction count pipeline 1
32	Pulse	Core 4	Instruction cache miss
33	Pulse	Core 4	Instruction TLB miss
34	Pulse	Core 4	Data cache L1 miss
35	Pulse	Core 4	Data TLB miss
36	Pulse	Core 4	Branch predictor miss
37	Pulse	Core 5	Instruction count pipeline 0
38	Pulse	Core 5	Instruction count pipeline 1
Continued on next page			

Table 2.1 – continued from previous page

Index	Type	Source	Description
39	Pulse	Core 5	Instruction cache miss
40	Pulse	Core 5	Instruction TLB miss
41	Pulse	Core 5	Data cache L1 miss
42	Pulse	Core 5	Data TLB miss
43	Pulse	Core 5	Branch predictor miss
44	CCS AHB	-	Agressor C1 Victim C0
45	CCS AHB	-	Agressor C2 Victim C0
46	CCS AHB	-	Agressor C3 Victim C0
47	CCS AHB	-	Agressor C4 Victim C0
48	CCS AHB	-	Agressor C5 Victim C0
49	CCS AHB	-	Agressor C0 Victim C1
50	CCS AHB	-	Agressor C2 Victim C1
51	CCS AHB	-	Agressor C3 Victim C1
52	CCS AHB	-	Agressor C4 Victim C1
53	CCS AHB	-	Agressor C5 Victim C1
54	CCS AHB	-	Agressor C0 Victim C2
55	CCS AHB	-	Agressor C1 Victim C2
56	CCS AHB	-	Agressor C3 Victim C2
57	CCS AHB	-	Agressor C4 Victim C2
58	CCS AHB	-	Agressor C5 Victim C2
59	CCS AHB	-	Agressor C0 Victim C3
60	CCS AHB	-	Agressor C1 Victim C3
61	CCS AHB	-	Agressor C2 Victim C3
62	CCS AHB	-	Agressor C4 Victim C3
63	CCS AHB	-	Agressor C5 Victim C3
64	CCS AHB	-	Agressor C0 Victim C4
65	CCS AHB	-	Agressor C1 Victim C4
66	CCS AHB	-	Agressor C2 Victim C4
67	CCS AHB	-	Agressor C3 Victim C4
68	CCS AHB	-	Agressor C5 Victim C4
69	CCS AHB	-	Agressor C0 Victim C5
70	CCS AHB	-	Agressor C1 Victim C5
71	CCS AHB	-	Agressor C2 Victim C5
72	CCS AHB	-	Agressor C3 Victim C5
73	CCS AHB	-	Agressor C4 Victim C5
74	CCS AXI	Write	Agressor MQ1 Victim MQ0
75	CCS AXI	Write	Agressor MQ2 Victim MQ0
76	CCS AXI	Write	Agressor MQ3 Victim MQ0
77	CCS AXI	Write	Agressor MQ4 Victim MQ0
78	CCS AXI	Write	Agressor MQ5 Victim MQ0
Continued on next page			

Table 2.1 – continued from previous page

Index	Type	Source	Description
79	CCS AXI	Write	Agressor MQ6 Victim MQ0
80	CCS AXI	Write	Agressor MQ7 Victim MQ0
81	CCS AXI	Write	Agressor MQ8 Victim MQ0
82	CCS AXI	Write	Agressor MQ9 Victim MQ0
83	CCS AXI	Write	Agressor MQ10 Victim MQ0
84	CCS AXI	Write	Agressor MQ11 Victim MQ0
85	CCS AXI	Write	Agressor MQ12 Victim MQ0
86	CCS AXI	Write	Agressor MQ13 Victim MQ0
87	CCS AXI	Write	Agressor MQ14 Victim MQ0
88	CCS AXI	Write	Agressor MQ0 Victim MQ1
89	CCS AXI	Write	Agressor MQ2 Victim MQ1
90	CCS AXI	Write	Agressor MQ3 Victim MQ1
91	CCS AXI	Write	Agressor MQ4 Victim MQ1
92	CCS AXI	Write	Agressor MQ5 Victim MQ1
93	CCS AXI	Write	Agressor MQ6 Victim MQ1
94	CCS AXI	Write	Agressor MQ7 Victim MQ1
95	CCS AXI	Write	Agressor MQ8 Victim MQ1
96	CCS AXI	Write	Agressor MQ9 Victim MQ1
97	CCS AXI	Write	Agressor MQ10 Victim MQ1
98	CCS AXI	Write	Agressor MQ11 Victim MQ1
99	CCS AXI	Write	Agressor MQ12 Victim MQ1
100	CCS AXI	Write	Agressor MQ13 Victim MQ1
101	CCS AXI	Write	Agressor MQ14 Victim MQ1
102	CCS AXI	Write	Agressor MQ0 Victim MQ2
103	CCS AXI	Write	Agressor MQ1 Victim MQ2
104	CCS AXI	Write	Agressor MQ3 Victim MQ2
105	CCS AXI	Write	Agressor MQ4 Victim MQ2
106	CCS AXI	Write	Agressor MQ5 Victim MQ2
107	CCS AXI	Write	Agressor MQ6 Victim MQ2
108	CCS AXI	Write	Agressor MQ7 Victim MQ2
109	CCS AXI	Write	Agressor MQ8 Victim MQ2
110	CCS AXI	Write	Agressor MQ9 Victim MQ2
111	CCS AXI	Write	Agressor MQ10 Victim MQ2
112	CCS AXI	Write	Agressor MQ11 Victim MQ2
113	CCS AXI	Write	Agressor MQ12 Victim MQ2
114	CCS AXI	Write	Agressor MQ13 Victim MQ2
115	CCS AXI	Write	Agressor MQ14 Victim MQ2
116	CCS AXI	Write	Agressor MQ0 Victim MQ3
117	CCS AXI	Write	Agressor MQ1 Victim MQ3
118	CCS AXI	Write	Agressor MQ2 Victim MQ3
Continued on next page			

Table 2.1 – continued from previous page

Index	Type	Source	Description
119	CCS AXI	Write	Agressor MQ4 Victim MQ3
120	CCS AXI	Write	Agressor MQ5 Victim MQ3
121	CCS AXI	Write	Agressor MQ6 Victim MQ3
122	CCS AXI	Write	Agressor MQ7 Victim MQ3
123	CCS AXI	Write	Agressor MQ8 Victim MQ3
124	CCS AXI	Write	Agressor MQ9 Victim MQ3
125	CCS AXI	Write	Agressor MQ10 Victim MQ3
126	CCS AXI	Write	Agressor MQ11 Victim MQ3
127	CCS AXI	Write	Agressor MQ12 Victim MQ3
128	CCS AXI	Write	Agressor MQ13 Victim MQ3
129	CCS AXI	Write	Agressor MQ14 Victim MQ3
130	CCS AXI	Write	Agressor MQ0 Victim MQ4
131	CCS AXI	Write	Agressor MQ1 Victim MQ4
132	CCS AXI	Write	Agressor MQ2 Victim MQ4
133	CCS AXI	Write	Agressor MQ3 Victim MQ4
134	CCS AXI	Write	Agressor MQ5 Victim MQ4
135	CCS AXI	Write	Agressor MQ6 Victim MQ4
136	CCS AXI	Write	Agressor MQ7 Victim MQ4
137	CCS AXI	Write	Agressor MQ8 Victim MQ4
138	CCS AXI	Write	Agressor MQ9 Victim MQ4
139	CCS AXI	Write	Agressor MQ10 Victim MQ4
140	CCS AXI	Write	Agressor MQ11 Victim MQ4
141	CCS AXI	Write	Agressor MQ12 Victim MQ4
142	CCS AXI	Write	Agressor MQ13 Victim MQ4
143	CCS AXI	Write	Agressor MQ14 Victim MQ4
144	CCS AXI	Write	Agressor MQ0 Victim MQ5
145	CCS AXI	Write	Agressor MQ1 Victim MQ5
146	CCS AXI	Write	Agressor MQ2 Victim MQ5
147	CCS AXI	Write	Agressor MQ3 Victim MQ5
148	CCS AXI	Write	Agressor MQ4 Victim MQ5
149	CCS AXI	Write	Agressor MQ6 Victim MQ5
150	CCS AXI	Write	Agressor MQ7 Victim MQ5
151	CCS AXI	Write	Agressor MQ8 Victim MQ5
152	CCS AXI	Write	Agressor MQ9 Victim MQ5
153	CCS AXI	Write	Agressor MQ10 Victim MQ5
154	CCS AXI	Write	Agressor MQ11 Victim MQ5
155	CCS AXI	Write	Agressor MQ12 Victim MQ5
156	CCS AXI	Write	Agressor MQ13 Victim MQ5
157	CCS AXI	Write	Agressor MQ14 Victim MQ5
158	CCS AXI	Read	Agressor MQ1 Victim MQ0
Continued on next page			

Table 2.1 – continued from previous page

Index	Type	Source	Description
159	CCS AXI	Read	Agressor MQ2 Victim MQ0
160	CCS AXI	Read	Agressor MQ3 Victim MQ0
161	CCS AXI	Read	Agressor MQ4 Victim MQ0
162	CCS AXI	Read	Agressor MQ5 Victim MQ0
163	CCS AXI	Read	Agressor MQ6 Victim MQ0
164	CCS AXI	Read	Agressor MQ7 Victim MQ0
165	CCS AXI	Read	Agressor MQ8 Victim MQ0
166	CCS AXI	Read	Agressor MQ9 Victim MQ0
167	CCS AXI	Read	Agressor MQ10 Victim MQ0
168	CCS AXI	Read	Agressor MQ11 Victim MQ0
169	CCS AXI	Read	Agressor MQ12 Victim MQ0
170	CCS AXI	Read	Agressor MQ13 Victim MQ0
171	CCS AXI	Read	Agressor MQ14 Victim MQ0
172	CCS AXI	Read	Agressor MQ0 Victim MQ1
173	CCS AXI	Read	Agressor MQ2 Victim MQ1
174	CCS AXI	Read	Agressor MQ3 Victim MQ1
175	CCS AXI	Read	Agressor MQ4 Victim MQ1
176	CCS AXI	Read	Agressor MQ5 Victim MQ1
177	CCS AXI	Read	Agressor MQ6 Victim MQ1
178	CCS AXI	Read	Agressor MQ7 Victim MQ1
179	CCS AXI	Read	Agressor MQ8 Victim MQ1
180	CCS AXI	Read	Agressor MQ9 Victim MQ1
181	CCS AXI	Read	Agressor MQ10 Victim MQ1
182	CCS AXI	Read	Agressor MQ11 Victim MQ1
183	CCS AXI	Read	Agressor MQ12 Victim MQ1
184	CCS AXI	Read	Agressor MQ13 Victim MQ1
185	CCS AXI	Read	Agressor MQ14 Victim MQ1
186	CCS AXI	Read	Agressor MQ0 Victim MQ2
187	CCS AXI	Read	Agressor MQ1 Victim MQ2
188	CCS AXI	Read	Agressor MQ3 Victim MQ2
189	CCS AXI	Read	Agressor MQ4 Victim MQ2
190	CCS AXI	Read	Agressor MQ5 Victim MQ2
191	CCS AXI	Read	Agressor MQ6 Victim MQ2
192	CCS AXI	Read	Agressor MQ7 Victim MQ2
193	CCS AXI	Read	Agressor MQ8 Victim MQ2
194	CCS AXI	Read	Agressor MQ9 Victim MQ2
195	CCS AXI	Read	Agressor MQ10 Victim MQ2
196	CCS AXI	Read	Agressor MQ11 Victim MQ2
197	CCS AXI	Read	Agressor MQ12 Victim MQ2
198	CCS AXI	Read	Agressor MQ13 Victim MQ2
Continued on next page			

Table 2.1 – continued from previous page

Index	Type	Source	Description
199	CCS AXI	Read	Agressor MQ14 Victim MQ2
200	CCS AXI	Read	Agressor MQ0 Victim MQ3
201	CCS AXI	Read	Agressor MQ1 Victim MQ3
202	CCS AXI	Read	Agressor MQ2 Victim MQ3
203	CCS AXI	Read	Agressor MQ4 Victim MQ3
204	CCS AXI	Read	Agressor MQ5 Victim MQ3
205	CCS AXI	Read	Agressor MQ6 Victim MQ3
206	CCS AXI	Read	Agressor MQ7 Victim MQ3
207	CCS AXI	Read	Agressor MQ8 Victim MQ3
208	CCS AXI	Read	Agressor MQ9 Victim MQ3
209	CCS AXI	Read	Agressor MQ10 Victim MQ3
210	CCS AXI	Read	Agressor MQ11 Victim MQ3
211	CCS AXI	Read	Agressor MQ12 Victim MQ3
212	CCS AXI	Read	Agressor MQ13 Victim MQ3
213	CCS AXI	Read	Agressor MQ14 Victim MQ3
214	CCS AXI	Read	Agressor MQ0 Victim MQ4
215	CCS AXI	Read	Agressor MQ1 Victim MQ4
216	CCS AXI	Read	Agressor MQ2 Victim MQ4
217	CCS AXI	Read	Agressor MQ3 Victim MQ4
218	CCS AXI	Read	Agressor MQ5 Victim MQ4
219	CCS AXI	Read	Agressor MQ6 Victim MQ4
220	CCS AXI	Read	Agressor MQ7 Victim MQ4
221	CCS AXI	Read	Agressor MQ8 Victim MQ4
222	CCS AXI	Read	Agressor MQ9 Victim MQ4
223	CCS AXI	Read	Agressor MQ10 Victim MQ4
224	CCS AXI	Read	Agressor MQ11 Victim MQ4
225	CCS AXI	Read	Agressor MQ12 Victim MQ4
226	CCS AXI	Read	Agressor MQ13 Victim MQ4
227	CCS AXI	Read	Agressor MQ14 Victim MQ4
228	CCS AXI	Read	Agressor MQ0 Victim MQ5
229	CCS AXI	Read	Agressor MQ1 Victim MQ5
230	CCS AXI	Read	Agressor MQ2 Victim MQ5
231	CCS AXI	Read	Agressor MQ3 Victim MQ5
232	CCS AXI	Read	Agressor MQ4 Victim MQ5
233	CCS AXI	Read	Agressor MQ6 Victim MQ5
234	CCS AXI	Read	Agressor MQ7 Victim MQ5
235	CCS AXI	Read	Agressor MQ8 Victim MQ5
236	CCS AXI	Read	Agressor MQ9 Victim MQ5
237	CCS AXI	Read	Agressor MQ10 Victim MQ5
238	CCS AXI	Read	Agressor MQ11 Victim MQ5
Continued on next page			

Table 2.1 – continued from previous page

Index	Type	Source	Description
239	CCS AXI	Read	Agressor MQ12 Victim MQ5
240	CCS AXI	Read	Agressor MQ13 Victim MQ5
241	CCS AXI	Read	Agressor MQ14 Victim MQ5
242	CCS AHB	-	Aggressor others Victim C0
243	-	-	Filler signal, constant 0
244	-	-	Filler signal, constant 0
245	-	-	Filler signal, constant 0
246	-	-	Filler signal, constant 0
247	-	-	Filler signal, constant 0
248	CCS AHB	-	Generic, Aggressor 0 Victim 0
249	CCS AHB	-	Generic, Aggressor 1 Victim 0
250	CCS AHB	-	Generic, Aggressor 2 Victim 0
251	CCS AHB	-	Generic, Aggressor 3 Victim 0
252	CCS AHB	-	Generic, Aggressor 4 Victim 0
253	CCS AHB	-	Generic, Aggressor 5 Victim 0
254	CCS AHB	-	Residual, Victim 0
255	-	-	Filler signal, constant 0

Table 2.2: Mapping from Master qos (MQ) to IPs on GPL release.

MQ	Bus of origin	Bus ID	Description
0	AHB	0	NOEL-V CORE
1	AHB	1	NOEL-V CORE
2	AHB	2	NOEL-V CORE
3	AHB	3	NOEL-V CORE
4	AHB	4	NOEL-V CORE
5	AHB	5	NOEL-V CORE
6	AHB	6	GR Ethernet MAC
7	AHB	7	GRDMAC2 DMA Controller
8	AHB	8	AHB Debug UART
9	AHB	9	JTAG Debug Link
10	AHB	10	EDCL master interface
11	-	-	Unused
12	-	-	Unused
13	-	-	Unused
14	-	-	Unused

Table 2.3: Mapping from Master qos (MQ) to IPs on non-GPL release

MQ	Bus of origin	Bus ID	Description
0	AHB	0	NOEL-V CORE
1	AHB	1	NOEL-V CORE
2	AHB	2	NOEL-V CORE
3	AHB	3	NOEL-V CORE
4	AHB	4	NOEL-V CORE
5	AHB	5	NOEL-V CORE
6	AHB	6	GR Ethernet MAC
7	AHB	7	GRSPW2 SpaceWire
8	AHB	8	GRSPW2 SpaceWire
9	AHB	9	CAN-FD Controller with DMA
10	AHB	10	CAN-FD Controller with DMA
11	AHB	11	GRDMAC2 DMA Controller
12	AHB	12	AHB Debug UART
13	AHB	13	JTAG Debug Link
14	AHB	14	EDCL master interface

Signals labeled as *debug* are fix inputs that can be used to test hardware or software with known inputs. **Event 0** (fix '1') can be used to measure the **number of elapsed cycles**.

Signals of type *CCS* can be used to compute the total contention cycle stack of the system. These signals become high at the first rising edge of the clock after a given condition or event has been detected. They remain active until the condition or event that they are measuring becomes low. This behavior allows measuring the length of clock cycles. When generating CCS signals, the user must consider if they want to allow back to back events and generate the input signals accordingly at RTL level.

The generic and residual contention metrics have the following meanings:

- **Generic contention:** The previous request has been served, but has left the slave in an unready state, making this core's request be delayed this cycle. We assume that the slave would be ready if the previous master had not accessed it.
- **Residual contention:** We observed that, when using the gaisler L2-lite cache, if a htrans "10"(NONSEQ) happens after a non-idle transaction, the response to the current master (the one that issued the "10") transaction is delayed by one cycle. THIS IS GAISLER L2 LITE specific, thus, not portable

2.3 MAIN CONFIGURATION AND SELF-TEST

Reset and enable of overflow and regular counters can be performed with register 2.1. All signals are active high.

Self-test mode allows to bypass the input events from the crossbar and instead use a specific

input pattern where signals are constant. This mode can be used for debugging. After the addition of the crossbar and debug inputs, there is a certain overlap. The same results can be achieved with the correct crossbar configuration. Nevertheless, it has been included in this release for compatibility.

These are the Self-test modes for each configuration value of the fied *selftest mode* in register 2.1:

- 0b00: Events depend on the crossbar. Self-test is disabled
- 0b01: All signals are set to 1.
- 0b10: All signals are set to 0.
- 0b11: Signal 0 is set to 1. The remaining signals are set to 0.

Register 2.1: BASE CONFIGURATION REGISTER (0x000)

Selftest mode		Reserved					Overflow softreset				Overflow enable				General Softreset				General Enable				
31	30	28					5					3	2	1	0								
00		x										0	0	0	0	Reset value							

2.4 CROSSBAR

This feature allows routing any of the input signals of table 2.1 or 2.2 into any of the 24 counters of the SafeSU. Each one of the counters has a 5-bit configuration value for the 32 input version of the crossbar, 7-bit for the 128 version and 8-bit for the 256 version. For simplicity this section describes the 32 entry variant of the crossbar. Consider that the width of the configuration field is $\log_2(N_{inputs})$ and the total amount of configuration registers $\log_2(N_{inputs}) * N_{inputs}$.

Configuration values are stored in registers 2.2, 2.3, 2.4, 2.5. All the configuration values are consecutive. Thus some values may have configuration bits in two consecutive memory addresses. Examples of this are Output 6, 12, 19 in our current configuration. As a consequence, the previous outputs may require two writes to configure the desired input signal.

Configuration fields match one to one with the internal counters. So the field *output 0* matches with *counter 0*, *output 1* with *counter 1* and so on.

As a usage example, suppose the user wants to route the signal *pmu_events(0).icnt(0)* to the internal *counter 0*. The field *Output 0* of register 2.2 shall match the Index of the signal in table 2.1. In this case, the index is 2. After this configuration, the event count will be recorded in *counter 0*. The addresses for counter values are indicated in figure 2.1.

Register 2.2: CROSSBAR CONFIGURATION REGISTER 0 (0x0AC)

Output 6 [1:0]		Output 5		Output 4		Output 3		Output 2		Output 1		Output 0	
31	30	29	25	24	20	19	15	14	10	9	5	4	0
00		00		00		00		00		00		00	

Reset value

Register 2.3: CROSSBAR CONFIGURATION REGISTER 1 (0x0B0)

Output 12[3:0]		Output 11		Output 10		Output 9		Output 8		Output 7		Output 6 [4:2]	
31	28	27	23	22	18	17	13	12	8	7	3	2	0
00		00		00		00		00		00		00	

Reset value

Register 2.4: CROSSBAR CONFIGURATION REGISTER 2 (0x0B4)

Output 19 [0:0]		Output 18		Output 17		Output 16		Output 15		Output 14		Output 13		Output 12[4:4]	
31	30	26	25	21	20	16	15	11	10	6	5	1	0		
00		00		00		00		00		00		00		00	

Reset value

Register 2.5: CROSSBAR CONFIGURATION REGISTER 3 (0x0B8)

Reserved		Output 24		Output 23		Output 22		Output 21		Output 20		Output 19[4:1]	
31	29	28	24	23	19	18	14	13	9	8	4	3	0
x		00		00		00		00		00		00	

Reset value

Signal routing is important since some of the SafeSU features are only available at different crossbar outputs. Table 2.4 shows the available capabilities for each one of the outputs.

⁰RDC and MCCU signals are only available if the assigned core is synthesized.

Table 2.4: Crossbar outputs and SafeSU capabilities

Output	Counters	Overflow	MCCU	RDC
0	Yes	Yes	Core 0	Yes
1	Yes	Yes	Core 0	Yes
2	Yes	Yes	Core 1	Yes
3	Yes	Yes	Core 1	Yes
4	Yes	Yes	Core 2	Yes
5	Yes	Yes	Core 2	Yes
6	Yes	Yes	Core 3	Yes
7	Yes	Yes	Core 3	Yes
8	Yes	Yes	Core 4	Yes
9	Yes	Yes	Core 4	Yes
10	Yes	Yes	Core 5	Yes
11	Yes	Yes	Core 5	Yes
12	Yes	Yes	No	No
13	Yes	Yes	No	No
14	Yes	Yes	No	No
15	Yes	Yes	No	No
16	Yes	Yes	No	No
17	Yes	Yes	No	No
18	Yes	Yes	No	No
19	Yes	Yes	No	No
20	Yes	Yes	No	No
21	Yes	Yes	No	No
22	Yes	Yes	No	No
23	Yes	Yes	No	No

2.5 COUNTERS

The unit in the default configuration contains 24 counters, 32-bit each. Figures 2.1 and 2.2 indicate the memory address where each counter's value can be access. Counter values can be **read** or **written**, thus allowing to set the initial value of the counters.

Enable and reset is managed by the base configuration register 2.1.

Counters can overflow. In such a case, the count will wrap back to 0 and keep counting. Section 2.6 describes how to enable the overflow detection interrupts.

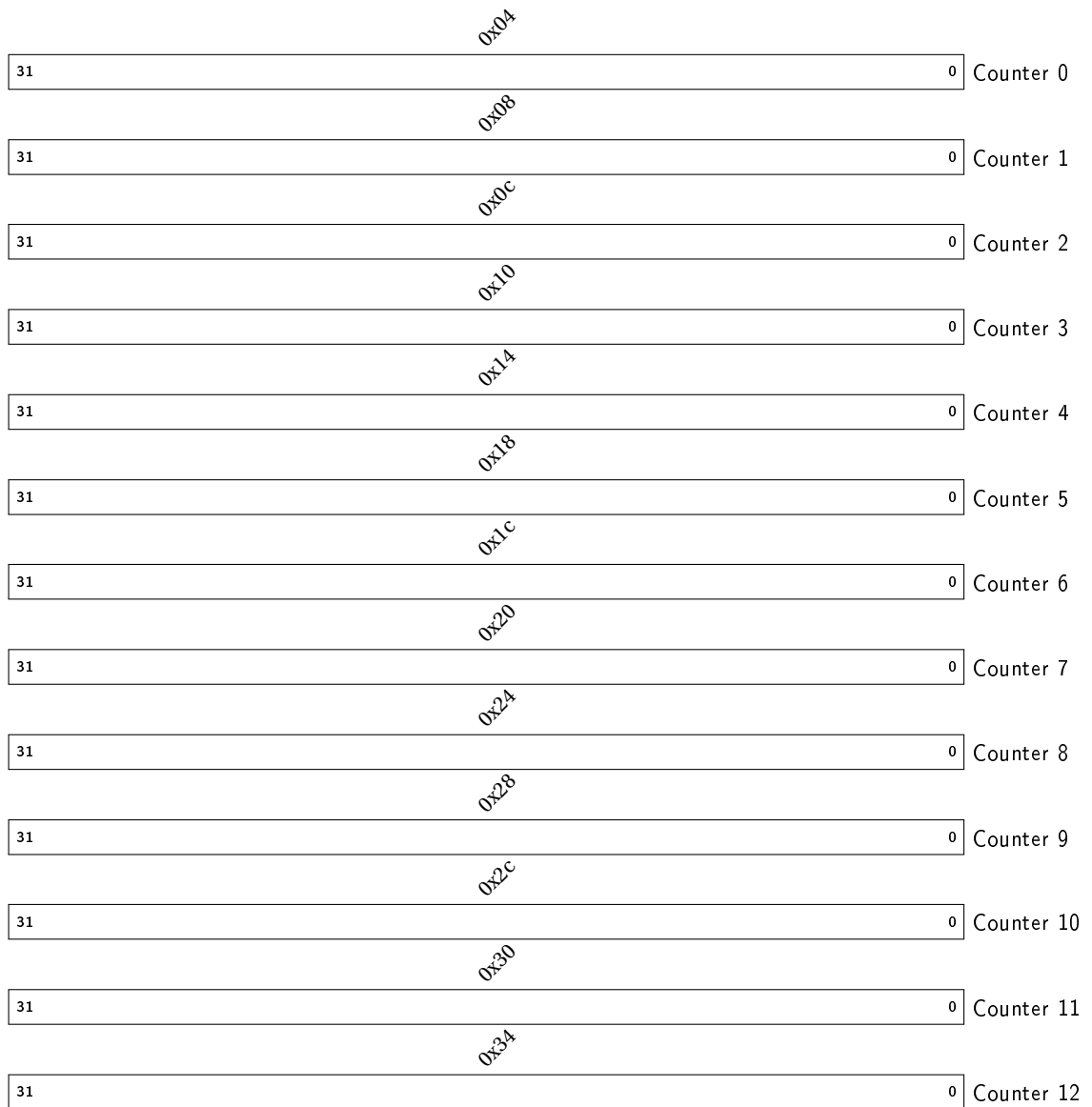


Figure 2.1: 0 to 12 Counter addresses



Figure 2.2: 12 to 24 Counter addresses

2.6 OVERFLOW

The user can enable overflow detection for each one of the counters in section 2.5. Enables are active high and individual for each counter, as indicated in register 2.6. If a counter with overflow detection active wraps over the maximum value, the corresponding bit of register 2.7 will become 1, and AHB interrupt number 6 will become active.

The default AHB interrupt mapping can be modified within the file *ahb_wrapper.vhd*.

Register 2.6: OVERFLOW INTERRUPT ENABLE MASK (0x064)

Reserved								Counter23	Counter22	Counter21	Counter20	Counter19	Counter18	Counter17	Counter16	Counter15	Counter14	Counter13	Counter12	Counter11	Counter10	Counter9	Counter8	Counter7	Counter6	Counter5	Counter4	Counter3	Counter2	Counter1	Counter0
31		24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0					
x								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset value

Register 2.7: OVERFLOW INTERRUPT VECTOR (0x068)

Reserved								Counter23	Counter22	Counter21	Counter20	Counter19	Counter18	Counter17	Counter16	Counter15	Counter14	Counter13	Counter12	Counter11	Counter10	Counter9	Counter8	Counter7	Counter6	Counter5	Counter4	Counter3	Counter2	Counter1	Counter0
31		24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0					
0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset value

2.7 MCCU

The Maximum-Contention Control Unit (MCCU) allows to monitor a subset of the input events and track the approximate contention that they will cause. Currently, events assigned to counters 0 to 11 can be used as inputs of the MCCU. Thanks to the crossbar, any of the SoC signals can be used by the MCCU. In this section we use the 4 core version as a reference. In all configurations one quota register is assigned to each core and one weight is assigned to each signal.

Register 2.8: MCCU MAIN CONFIGURATION FOR 6 CORE CONFIGURATIONS (0x074)

Enable Hardware Quota										Reserved										Soft reset RDC Enable RDC Update Quota Core 5 Update Quota Core 4 Update Quota Core 3 Update Quota Core 2 Update Quota Core 1 Soft reset MCCU Enable reset MCCU									
31	30									10									9	8	7	6	5	4	3	2	1	0	
0	x																		0	0	0	0	0	0	0	0	0	0	

Reset value

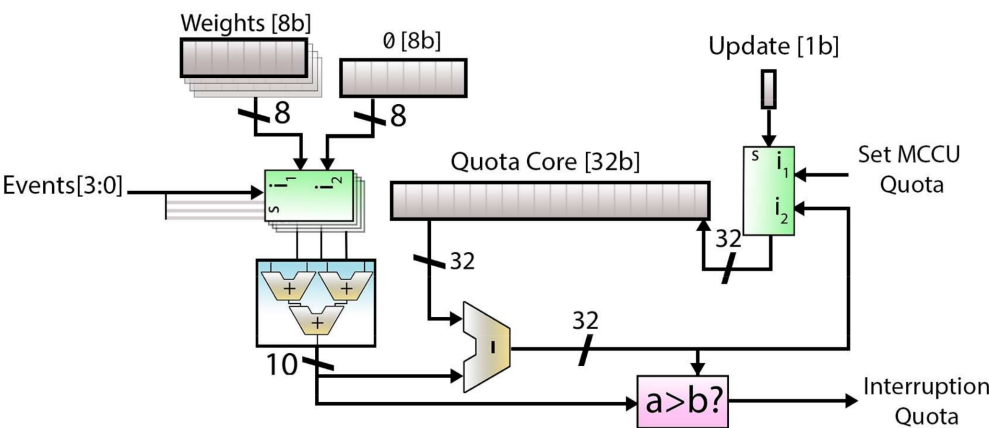


Figure 2.3: Block diagram MCCU mechanism for one core.

Figure 2.3 shows the internal elements required to monitor the quota consumption of one core, given four input events. When events are active, they pass the value assigned in the weight register 2.9 for the given signal to a series of adders. The addition is subtracted from the corresponding quota register 2.5. When the remaining quota is smaller than the cycle contention, an interrupt is triggered.

When the "Enable Hardware Quota" bit is toggled, this interrupt is routed to the AHB controller directly to block contending core access to the shared bus. Thus, disabling the interrupt lines and performing the decisions without software intervention.

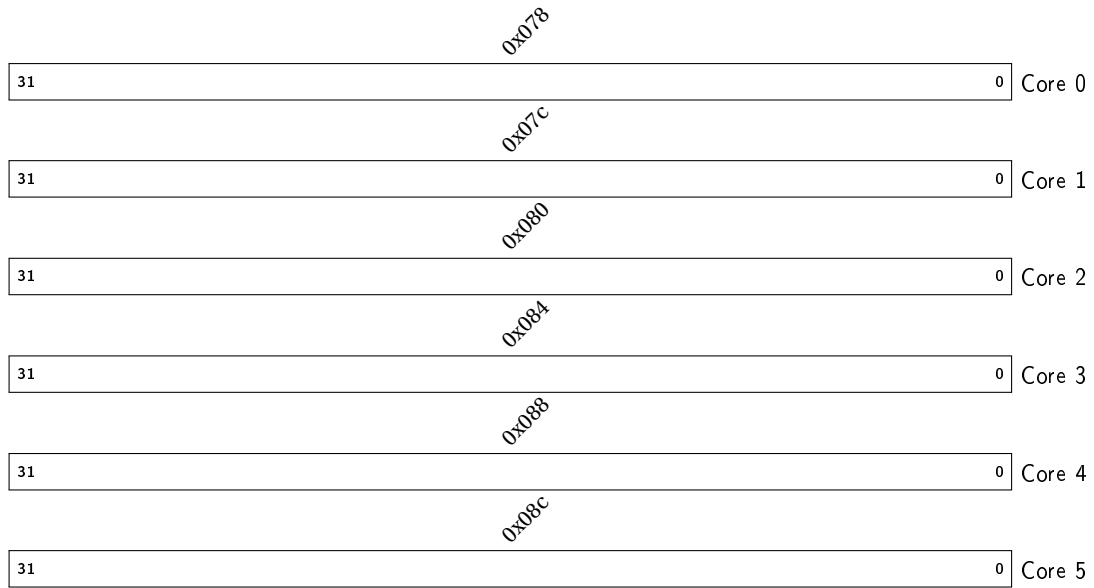


Figure 2.4: MCCU Quota limits for each core



Figure 2.5: MCCU Current remaning Quota for each core

In the current release, the MCCU can be reset and activated with the respective fields of register 2.8. The fields labeled as *Update Quota core* are used to update the available quota of each core (figure 2.5). While *Update Quota core* is high, the content is assigned to the avail-

able quota. Once released (low), the available quota can start to decrease if the MCCU is active. The current quota can be read while the unit is active.

In the current release, each core can monitor two input events. The MCCU module is parametric and more events can be provided in future releases. Table 2.4 shows the features available for each crossbar output. Under the column MCCU, you can see towards which core quota the event will be computed. The unit provides one interruption for each of the monitored cores.

Weights for each monitored event are registered in registers 2.9 and 2.11. Currently, each weight is an 8-bit field. Each input of the MCCU maps directly to the outputs of the crossbar. Thus the weight for the MCCU input 0 corresponds to the signal in crossbar output 0.

Register 2.9: MCCU EVENT WEIGHTS REGISTER 0 (SHARED WITH RDC) (0x0a8)

Input 3				Input 2				Input 1				Input 0				
31		24		23		16		15		8		7		0		
00				00				00				00				Reset value

Register 2.10: MCCU EVENT WEIGHTS REGISTER 1 (SHARED WITH RDC) (0x0ac)

Input 7				Input 6				Input 5				Input 4				
3124				2316				158				70				
00				00				00				00				Reset value

Register 2.11: MCCU EVENT WEIGHTS REGISTER 2 (SHARED WITH RDC) (0x0b0)

Input 11				Input 10				Input 9				Input 8				
31		24		23		16		15		8		7		0		
00				00				00				00				Reset value

2.8 RDC

The Request Duration Counter or RDC is comprised of a set of 8-bit counters and comparators that allow monitoring the length of a CCS signal, record the number of clock cycles of the longest pulse and compare such value with the defined weight.

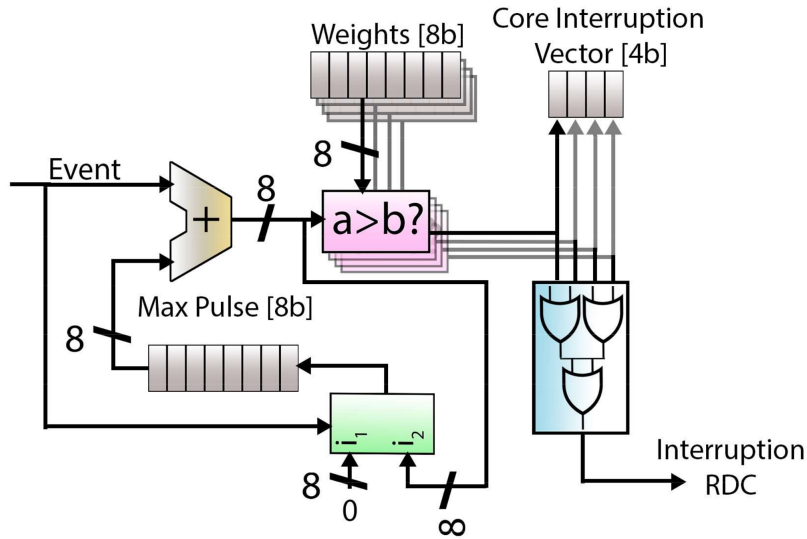


Figure 2.6: Block diagram RDC mechanism.

The current release provides monitoring for crossbar outputs 0 to 11 for the hexa-core. Note that the amount of registers changes with the amount of cores, and figures and addresses are representative of the 4 core configuration. The weights for each signal are shared with the MCCU and are stored in registers 2.13, 2.14 and 2.15. Weights are 8-bit fields. Counters have overflow protection, preventing the count from wrapping over the maximum value. The maximum value for each event (watermarks), are stored in registers 2.16, 2.17 and 2.18.

The RDC shares the main configuration register with the MCCU (register 2.8). Through this register, the unit can be reset and enabled through the corresponding fields. Such fields are active high signals.

The unit does provide access to the internal interrupt vector (register 2.12), but such information is redundant and may be removed in future releases. Given the current watermarks and assigned weights, the events responsible for the interrupt can be identified.

Weights not configured or with value 0 will not raise interruptions.

Register 2.12: RDC INTERRUPT VECTOR (0x0b4)

Reserved					Core 5	Core 4	Core 3	Core 2	Core 1	Core 0	
31					4	3	3	3	2	1	0
x						00	00	00	00	00	00

Reset value

Register 2.13: RDC EVENT WEIGHTS REGISTER 0 (SHARED WITH MCCU) (0x0a8)

Input 3				Input 2				Input 1				Input 0			
31	24	23	16	15	8	7	0								
00				00				00				00			

Reset value

Register 2.14: RDC EVENT WEIGHTS REGISTER 1 (SHARED WITH MCCU) (0x0ac)

Input 7				Input 6				Input 5				Input 4			
31	24	23	16	15	8	7	0								
00				00				00				00			

Reset value

Register 2.15: RDC EVENT WEIGHTS REGISTER 2 (SHARED WITH MCCU) (0x0b0)

Input 11				Input 10				Input 9				Input 8			
31	24	23	16	15	8	7	0								
00				00				00				00			

Reset value

Register 2.16: RDC WATERMARK REGISTER 0 (0x0b8)

Input 3				Input 2				Input 1				Input 0			
31	24	23	16	15	8	7	0								
00				00				00				00			

Reset value

Register 2.17: RDC WATERMARK REGISTER 1 (0x0bc)

Input 7				Input 6				Input 5				Input 4			
31	24	23	16	15	8	7	0								
00				00				00				00			

Reset value

Register 2.18: RDC WATERMARK REGISTER 2 (0x0c0)

Input 11				Input 10				Input 9				Input 8			
31	24	23	16	15	8	7	0								
00				00				00				00			

Reset value

2.9 SOFTWARE SUPPORT

The unit can be configured by the user at a low level following the description of previous sections and the documents associated for each unit. In addition we provide a small bare-metal driver under the *drivers* directory of the SafeSU repository.

Currently 3 configurations are provided. *4-core* and *6-core* folders have the driver for default configurations. *6-core-256e* folder has a driver for an hexacore with 256 entries on the crossbar.

The drivers are composed of three files.

- *pmu_vars.h* : Defines a set of constants with the RTL parameters that generated the unit. Such constants allow the reusing of functions among hardware configurations.
- *pmu_hw.h*: Defines the memory position of the SafeSU within the memory map of the SoC. It also contains the prototypes of each function of the driver.
- *pmu_hw.c*: Contains the definition of each of the functions.

Given the current development status, the driver is not autogenerated, and **modifications to the RTL may require manual changes on the previous files.**

3 CONFIGURATION OPTIONS

Table 3.1 shows the configuration parameters exposed by *ahb_wrapper.vhd*. Given the current development status, changes to the configuration options may require manual modifications to internal modules and software drivers. Future releases will expose parameters to enable individual SafeSU features and allow for more flexibility.

Table 3.1: Configuration options (VHDL ports)

Generic	Function	Allowed range	Default
HADDR	AHB base address	0 to 16#fff#	0
HMASK	AHB address mask	0 to 16#fff#	16#fff#
N_REGS	Total of accessible registers	2 to 64	43
SafeSU_COUNTERS	Number of generic counters. Same as crossbar outputs	1 to 32	24
N_SOC_EV	SoC signals. Inputs to the crossbar	1 to 64	32
REG_WIDTH	Size of registers and counters	32 or 64	32

4 SIGNAL DESCRIPTIONS

Table 4.1 shows the interface of the core (VHDL ports).

Table 4.1: Signal descriptions (VHDL ports)

Signal name	Field	Type	Function	Active
RST		Input	Reset	Low
CLK		Input	AHB master bus clock	-
SafeSU_EVENTS		Input	Input for regular SoC events	-
CCS_CONTENTION		Input	Input for contention cycle stack signals that measure contention	-
CCS_LATENCY		Input	Input for contention cycle stack signals that measure access latency	-
AHBSI	*	Input	AHB slave input signals	-
AHBSO	*	Output	AHB slave output signals, includes interrupts	-

5 LIBRARY DEPENDENCES

Table 5.1 shows the libraries used when instantiating the core (VHDL libraries).

Table 5.1: Library dependencies

Library	Package	Imported units	Description
IEEE	std_logic_1164	Types	Standard logic types
GRLIB	amba	Signals	AMBA signal definitions
GRLIB	config	Types	Amba P&P types
GRLIB	devices	Types	Device names and vendors
GRLIB	stdlib	All	Common VHDL functions
GAISLER	noelv	Signals	Counter vectors and types
BSC	pmu_module	Instances and signals	Instances and signal definitions for the SafeSU

6 INSTANTIATION

An example design is provided in the context of SELENE and De-RISC project. Integration examples of earlier releases of the unit along LEON3MP can be provided under demand.

Listing 1: SafeSU instance example for gpp_sys

```
-- Include BSC library
library bsc;
use bsc.pmu_module.all;

--Provide a non-overlapping hsidx
constant hsidx_pmu : integer := 6;

--Update the hsidx of the next slave
```

```

constant nextslv      : integer := hsidx_pmu + 1;

--Declare events and signals of interest. They may change
--for each use case. Route such events to pmu_events,
--ccs_contention and ccs_latency ports

--Instance of the unit

PMU_inst : ahb_wrapper
generic map(
ncpu    => CFG_NCPU ,
hindex  => hsidx_pmu ,
haddr   => 16#801# ,
hmask   => 16#FFF#
)
port map(
rst      => rstn ,
clk      => clk ,
pmu_events    => pmu_events ,
ccs_contention    => ccs_contention ,
ccs_latency    => ccs_latency ,
ahbsi          => ahbsi ,
ahbso          => ahbso(hsidx_pmu));

```

Given the development status of the module and drivers it is recommended to not modify the internal parameters of the module.